

INFO90010 Technology Innovation Project

Assessment Task 1 – Group Project Proposal

You should start working on this document with your project team members during the week 2 workshop. Submit the final version by the start of week 3 (09:00am, Monday 16 March).*

You will receive feedback from your teaching team in the week 3 lecture and workshop.

***IMPORTANT:** Before submitting your completed proposal, you will need to add your team-mates to a group on the LMS. The groups have been created and labelled with a different letter of the alphabet (A-G). If you want to, you can edit the group name to something more personal. If you have any difficulty with this step, please contact the subject coordinator.

Please select a contact person of the group, who will communicate with the teaching team for common issues of the group on behalf of the team members.

Group details

Group Name		Group B	
Group Members Names, email addresses, and roles in project team			
	Name	Email address	Major role and responsibility (R&R) (e.g., contact person; user/customer research; design lead; software developer; project manager)
Group member 1:	Yang-Chun Lin	yangchunlin@student.unimelb.edu.au	Contact Person, Product Owner, Team Leader
Group member 2:	Hui-Chun Lee	huichunlee@student.unimelb.edu.au	Software Developer, User Research
Group member 3:	Shu-Han Yang	shuhyang3@student.unimelb.edu.au	Software Developer, Testing Developer
Group member 4:	Yuqi Chen	ychen4176@student.unimelb.edu.au	Software Developer, Design Lead

PART A - Brainstorming about the topic

Project Topic What is the problem you plan to investigate and innovate for? Why is this problem important?	Urban residents or international students frequently encounter situations where they need temporary access to tools, resources, or in-person assistance but lack convenient ways to obtain them locally. Examples include borrowing tools for furniture assembly or attending rental inspections. Purchasing rarely used items or looking for formal services can be costly and inefficient. At the same time, many useful resources and people who'd like to help others are available within local communities. This problem is relevant in cities with large student populations, like Melbourne and Sydney. International students often have limited social networks and may lack access to nearby support systems. Research has shown that international students usually experience social isolation and reduced local support networks when adapting to a new environment (Smith & Khawaja, 2011). Besides, the growth of the sharing economy proves that the interest in community-based access to resources has been increasing (Botsman & Rogers, 2010; Frenken & Schor, 2017). However, many existing sharing platforms focus on commercial transactions rather than lightweight, hyper-local exchanges of help between community members. Current solutions like Facebook Marketplace may not address small, informal interactions effectively and efficiently, such as borrowing a screwdriver or getting help with house moving. These interactions require trust, convenience, and strong local connections. Our project explores the potential for a hyper-local digital platform that enables community members to request or provide small tasks and resources nearby. By incorporating reputation systems and a gamified reward mechanism, the platform may attract more participants and strengthen trust within local communities. Innovative technology-based solutions could help reduce resource waste, improve access to informal support networks among residents living in the nearby areas or students.
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People (users & stakeholders) Who are the main people affected by the problem you plan to innovate for? Who are your intended users? Who are the other stakeholders (i.e., other people affected by, or involved in, the design context)? What do you think might be the concerns or priorities of your intended users and stakeholders?	Intended Users The primary users of this platform are urban students, particularly international students, who often face difficulties when dealing with small everyday tasks that require tools, equipment, or assistance. Examples include borrowing tools such as screwdrivers or ladders, moving small furniture or appliances, or getting quick help with household or technical issues. International students may especially benefit from this platform because they often lack nearby family members or established local networks who could help with these tasks. Concerns and priorities of intended users 1. Convenience: Users want a quick and easy way to find nearby help without needing to contact multiple people or hire professional services. 2. Speed of response: Since many requests involve small tasks, users expect help to be available relatively quickly. 3. Safety and trust: Users may feel hesitant interacting with strangers. Therefore, the platform should provide mechanisms such as verified accounts or user ratings. 4. Fair rewards or recognition: Users who offer help may expect small rewards, appreciation, or a sense of contribution to the community. Helpers Another important group of users are people willing to offer help. These users may enjoy helping others, earning small rewards, or building connections within their local community. Concerns and priorities of helpers 1. Proximity: They are more likely to accept requests that are close to their current location. 2. Clarity of the task: Helpers want clear descriptions of what the task involves and how long it might take. 3. Recognition or incentives: Some users may be motivated by rewards, reputation scores, or social recognition within the community. Other Stakeholders: Universities and student communities
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	Universities may benefit from platforms that improve student wellbeing and community support. Their concerns may include ensuring student safety, promoting positive community interaction, and preventing misuse of the platform. Local communities and residents Residents living near students may also interact with the platform as helpers or requesters. Their concerns may include privacy and security, avoiding nuisance requests, and maintaining respectful community behaviour. Development team The development team (the project group) is also a key stakeholder because they are responsible for designing and implementing the system. Their priorities include developing a functional prototype within the project timeframe, balancing innovation with technical feasibility, and ensuring the platform is simple and easy to use.
Tasks and goals What are the key tasks that people might be trying to achieve in this situation?	There are multiple scenarios where people might want to apply our product/service: • Request Immediate Assistance: A student moving into a new place may need a specific tool or physical help, but he doesn't want to buy one-time stuff • Building Social Connections: Users aiming to meet and connect with their neighbours through helpful/thoughtful actions • Delegated Community Tasks: Someone seeking people to attend house inspections or pick up something on their behalf when they are unavailable • Resource Sharing: Residents looking to lend or borrow items to reduce waste and save money

Technologies and prior innovations

What previous innovations have been introduced to address this problem? What technologies and tools are *currently in use* in this setting (that you are aware of)? [Tip: you can include any tools here, including paper-based resources] What do you believe are the limitations of existing tools?

People already use a mixture of digital and non-digital tools to solve small everyday problems in their local communities. When someone needs help, they often ask friends directly, send messages in messaging groups. Sometimes they simply borrow items from people they already know nearby. These approaches show that community members are generally willing to help each other and that there is an existing culture of informal sharing and mutual aid. However, these tools usually depend on existing social relationships. It is easy to ask someone you already know for help, but newcomers, students, or young renters may not feel comfortable asking neighbours they have never met before.

Several digital platforms have attempted to address parts of this problem. One example is **Nextdoor**, a neighbourhood-based social networking platform that allows residents to discuss local issues, ask for recommendations, and give away or sell items. However, the platform is quite broad and discussion-oriented. It is designed mainly for community conversations rather than for quick, small, time-sensitive requests between neighbours.

Another example is **Olio**, an application that enables people to share surplus food and household goods within their local area. But, its focus is primarily on redistributing surplus items rather than supporting the broader range of everyday neighbourly assistance.

There are also systems based on the **Library of Things**, where community members can borrow items such as kitchen appliances, tools, or electronics instead of purchasing them. This model promotes sustainability and reduces unnecessary consumption. While these systems work well for planned borrowing, they generally follow a structured lending library approach with scheduled borrowing periods. As a result, they are less suited for informal and spontaneous borrowing between neighbours, such as asking to use a screwdriver for thirty minutes.

From a technological perspective, many enabling technologies already exist that could support improved neighbourhood mutual aid systems. Most users today have access to smartphones, messaging applications, location services, and interactive digital maps. These technologies make it possible to develop web or mobile platforms that allow users to view requests and offers from nearby people in a map-based interface. Location-based features can help users identify nearby neighbours without requiring them to reveal their exact home addresses, which helps maintain privacy while still enabling local interaction.

Innovation expectations	Current solutions are mostly fragmented (like Facebook groups or Rednote posts) and lack real-time, location-based efficiency. Key pain points include the difficulty of finding help “quickly” and the lack of incentive for consistent engagement or to strengthen people’s connection or bonding. Potential directions for innovation: ● AI-Powered Request Routing (Gemini API): Instead of using a static list or having people manually check active users, we can use the Gemini API to perform NLP on user requests. For example, if a student asks for a screwdriver, the AI can automatically categorize the intent and proactively notify nearby neighbors who have “tools” listed in their profile, significantly reducing the time it takes to find help ● Gamified Social Engineering: Provide a rewarding mechanism by using AI to generate personalized “Community Quests”. By reaching milestones, like helping 5 neighbors, users can unlock unique skins or credit points for their digital identity, turning local altruism into an engaging, game-like experience ● Trust Layer & Safety: AI-driven content moderation can scan posts/pictures for safety. Multimodal AI features could be used to verify the condition of shared items through image analysis, ensuring the items are still working ● Map-Based Interface: An interactive map that lets users see “help flags” or “available resources” in their immediate vicinity, making the community’s needs visible
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PART B – PLANNING THE PROJECT

Investigating the topic What do you need to learn about the topic/problem you are investigating and innovating for? How will you go about this investigation?	Q1: Do students actually need a platform for small local help? Small tasks such as borrowing tools, carrying furniture, or fixing small appliances may occur frequently among students, but it is unclear how often these situations happen and how people currently solve them. Investigation approach: 1. Conduct a short survey among university students to understand: how often they need small help, what kinds of help are most common, whether they would use such a platform. 2. Interview international students, who may have less local support networks. 3. Analyse survey responses to identify common needs and user behaviour. Q2: What existing platforms already solve similar problems? There are existing platforms such as: Airtasker, TaskRabbit, Nextdoor. However, these platforms typically focus on paid services or social networking, rather than quick small-scale assistance. Investigation approach: 1. Analyse these platforms to understand: their main features, pricing models, limitations for small tasks. 2. Identify the gap in the market, particularly for micro-help within a close geographic area. Q3: What features are most important for users? The platform must provide useful features while remaining simple enough for a small team to develop. Investigation approach: 1. Review similar community platforms 2. Gather feedback from potential users about preferred features 3. Prioritise essential functions for a minimum viable product (MVP) Possible features to investigate include: map-based task discovery, task reward system, helper rating system, safety and identity verification.
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Milestones What are the key tasks you plan to complete before the Week 7 progress presentation, and by the end of semester? For each milestone, estimate the time required and align it with your team's R&R and available resources. How will you track progress and ensure you meet milestones?	Milestone 1 – Problem Research and Requirement(2 weeks) Key tasks: • Conduct student surveys and interviews • Analyse existing platforms • Identify key user needs • Define core features of the platform Deliverable: Problem analysis report and Initial feature list. Milestone 2 – System Design and Planning (1 week) Key tasks: • Design system architecture • Define database structure • Design user interface wireframes • Allocate roles and responsibilities among team members Deliverable: System architecture diagram, UI prototype design, and development plan. Milestone 3 – Prototype Development (Before Week 7) Key tasks: • Implement map interface • Develop request posting function • Display help requests on the map • Allow users to view request details Deliverable: functional prototype for progress presentation Milestone 4 – Feature Expansion and Testing(3-4 weeks) Key tasks: • implement task acceptance and completion functions
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	• develop user account system • test platform usability and stability • gather feedback from test users Deliverable: improved working prototype Milestone 5 – Final Refinement and Presentation (1–2 weeks) Key tasks: • fix technical issues • improve interface design • prepare final demonstration and presentation materials Deliverable: final prototype, and project presentation. Progress Tracking To ensure milestones are met, the team will: • hold weekly meetings to review progress • use project management tools such as GitHub • track task completion and assign responsibilities • update development progress regularly
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Risks and constraints What potential risks or constraints could delay or limit your project? Consider uncertainties that may impact your investigation or innovation. How will you mitigate these risks?	Risk 1 – Technical Complexity Implementing location-based services and map integration may be technically challenging for the team. Mitigation: • use existing APIs such as Google Maps or Mapbox • focus on a simplified MVP rather than full functionality • prioritise core features only Risk 2 – Safety and Trust Concerns Users may hesitate to interact with strangers when offering or requesting help. Mitigation: • require account registration • implement a basic user rating system • restrict testing to university students during the project Risk 3 – Low User Engagement If users do not actively post or accept tasks, the platform may appear inactive. Mitigation: • design a simple and intuitive interface • focus on a specific user group such as university students
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PART C - COMPLETE THE GROUP CONTRACT (SEE LMS)

Technology Innovation Project (INFO90010): Team Agreement

Date: 14/03/2026

Tutor: Mayura Manawadu

Group name: Group B

Group members: 4

GOALS: What are our team goals for this project? What do we want to accomplish? What skills do we want to develop or refine? Our main goal for this topic is to design a community-help platform for everyone. It allows people living nearby to request help, share resources, and connect their neighbours. Through this project, we aim to build both a website and mobile app that demonstrates the features of the idea, including the map-based interface, the needs and supply pages, and basic communication between users. Another important goal is to improve our technical skills. During this project, we might face a lot of technical problems. This will be a good opportunity to gain more experience in both frontend and backend development. The most worthy part is that we can learn how to find the problems in our tasks and learn how to solve them by ourselves. We also want to strengthen our teamwork and communication ability. Since this project requires collaboration across different tasks, effective collaboration will be essential to ensure the project progresses smoothly.
EXPECTATIONS: What do we expect of one another in regard to attendance at meetings, participation, frequency of communication, the quality of work, etc.? As a team, we expect everyone to be responsible, supportive, and actively involved in the project. Since this project requires collaboration, it is important that all team members communicate openly and participate in the development process. For meetings, we expect all members to attend meetings on time. Our plan is to have at least one meeting per week to discuss progress, share ideas, and assign tasks. If the project requires additional discussion, we may schedule an extra meeting when necessary. We prefer to meet in person on campus, because face-to-face communication makes discussion more efficient and won't be affected by the internet. However, if someone cannot come to campus due to special circumstances, they can join the meeting through Zoom online so that they can still participate. For communication, we hope that everyone can stay active in our WhatsApp group, which will be our main communication channel. If someone has an idea, question, or problem, we encourage them to share it with the team immediately instead of trying to solve everything alone. For participation, each team member should contribute to the project by completing assigned tasks on time, sharing feedback during discussion, and helping other team members when needed. Finally, regarding the quality of work, we expect all team members to aim for a standard that satisfies the whole team. We will review each other's work and provide suggestions for improvement so that the final result reflects the best effort of the entire team.
TEAM RULES: What rules do we agree on to help us meet our goals and expectations? • We agree to complete assigned tasks on time and take responsibility for our work. • We will respect each other's ideas and feedback during discussion. • We agree to communicate openly and regularly. • We should respect meeting times and attend meetings punctually. • We need to be friendly to each other.
COMMUNICATION & PROJECT MANAGEMENT: What tools will you use to communicate and manage tasks? How will project progress be recorded and communicated? For communication, we will mainly use WhatsApp as our primary communication channel. This allows team members to quickly contact and share ideas. It will also be used to arrange meeting times and send reminders about deadlines. For code management, technical material and meeting notes, we will use GitHub. GitHub can manage version control, and track changes made by different team members. For document collaboration, we will use Google Drive to store and edit shared documents.
ROLES & RESPONSIBILITIES (R&R): What roles will each team member take, and what is each person responsible for?

We agree that everyone will contribute to the technical development of the project, including both website and mobile application. However, each member will also take on additional responsibilities to help organise the project and ensure team works efficiently.

Leo will take the role of team leader and product owner. Leo will help guide the overall direction of the project, and ensure the project stays aligned with our original idea and goals.

Tammy will be responsible for documentation and project materials. This includes submit project files, maintaining files on Google Drive, and recording meeting notes. Tammy will ensure that our documentation is clear, complete, and submitted on time.

Bella will focus on the technical implementation and system design. This role involves paying close attention to the technical structure of the project.

Anita will focus on testing and quality assurance. She will help test the system's features, identify bugs or usability issues, and provide feedback to improve the platform before submission.

CONSEQUENCES: How will we address non-performance in regard to these goals, expectations, and agreed processes?

If a team member is not meeting the agreed goals, expectations, or responsibilities, we will first try to address the issue through open communication within the team. We believe that discussing the problem directly and respectfully can help identify the reason for the problem. If the problem continues after internal discussion, we will seek guidance from our tutor. If the issue still cannot be resolved, the team may need to report the situation to the subject coordinator. This step will only be taken if other solutions have not worked, and hope this can ensure that the project assessment remains fair for everyone in the team.

We share these goals and agree to these expectations, rules, and consequences.

Yang-Chun Lin

Team member 1 – write name

Shu-Han Yang

Team member 2 – write name

Yuqi Chen

Team member 3 – write name

Hui-Chun Lee

Team member 4 – write name

Declaration

We acknowledge the use of ChatGPT to check the grammar is correct and the sentences are fluent.

A full record of prompts and outputs is available upon request.